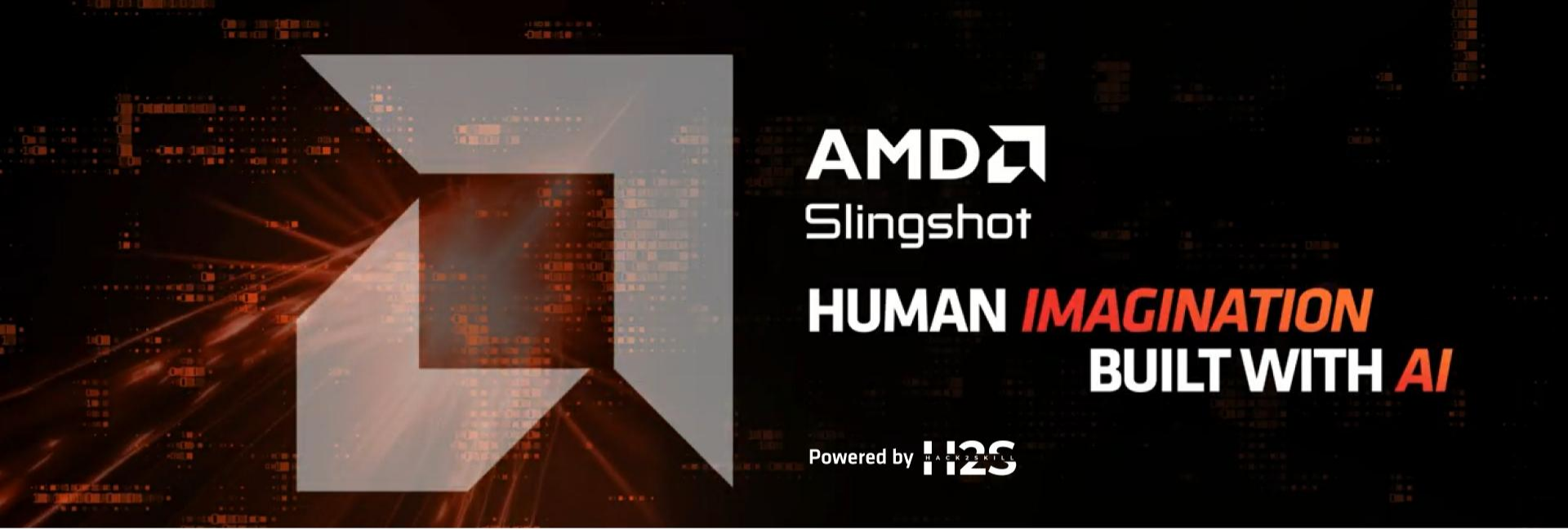

Team Details

- ➔ Team name - NeuralNova AI
- ➔ Team leader name - Y Naveen Raj Yadav
- ➔ Problem Statement - AI in Consumer Experiences

Brief about the idea

PersonaAI is a real-time AI-powered personalization engine designed to enhance digital consumer experiences. Unlike traditional systems that rely only on past purchase history, PersonaAI analyzes live behavioral signals such as browsing duration, click frequency, and spending patterns to dynamically classify users into behavioral personas.

Based on these personas, the system automatically adapts user interface layout, product recommendations, and promotional strategies in real time. This creates a highly personalized, intelligent, and adaptive consumer experience across e-commerce, OTT platforms, fintech apps, and digital retail ecosystems.

Opportunities

- ➔ How different is it from any of the other existing ideas?
- ➔ How will it be able to solve the problem?
- ➔ USP of the proposed solution

- Opportunities

• Rising demand for hyper-personalized digital experiences
• Growing AI adoption in retail and e-commerce
• Increasing competition among consumer platforms
• Need for higher customer retention and engagement

PersonaAI can be integrated into websites, mobile apps, OTT platforms, and fintech systems.

- How different is it from existing ideas?

Most current personalization systems:

• Use static recommendation algorithms
• Depend mainly on past purchase history
• Do not adapt UI dynamically
• Provide generic user experience

PersonaAI introduces:

• Real-time behavioral clustering
• Dynamic UI personalization
• Continuous persona reclassification
• Behavior-driven interface adaptation

It acts as a personalization engine rather than just a recommendation tool.

- How will it solve the problem?

Collects behavioral data (time spent, click intensity, spending patterns)

Processes data using machine learning clustering (KMeans)

Assigns users to dynamic persona categories

Modifies UI layout and recommendations accordingly

Continuously updates persona based on new interactions

This reduces decision fatigue and increases engagement.

- USP of the proposed solution

• Real-time adaptive personalization
• Behavior-driven clustering engine
• Cross-platform scalability
• Low-latency processing using AMD acceleration
• Industry-ready SaaS architecture

List of features offered by the solution

- Behavioral Data Collection Module
- KMeans-based User Clustering
- Persona Classification System
- Dynamic UI Adaptation Engine
- Intelligent Recommendation Module
- Behavioral Analytics Dashboard
- Scalable API Integration

Process flow diagram or Use-case diagram

User Interaction

↓

Behavior Data Collection

↓

Data Preprocessing

↓

KMeans Clustering Model

↓

Persona Assignment

↓

Dynamic UI & Recommendation Adjustment

↓

Continuous Feedback Loop

Wireframes/Mock diagrams of the proposed solution (optional)

1. Budget Persona
- Discount-first layout
- Price filters highlighted
- Offer banners prioritized

2. Premium Persona
- Premium product spotlight
- Exclusive deals highlighted
- Luxury UI theme

3. Fast Decision Persona
- Minimal layout
- Quick-buy options
- Top-rated items only

Architecture diagram of the proposed solution

Frontend Layer
• React.js User Interface

↓

API Layer
• FastAPI Backend

↓

AI Engine
• Python + Scikit-learn (KMeans Clustering)

↓

Database
• MongoDB (User behavioral data storage)

↓

Compute Layer
• AMD Ryzen / Radeon GPU acceleration for real-time inference

Technologies to be used in the solution

- Python
- Scikit-learn (KMeans)
- FastAPI
- React.js
- MongoDB
- GitHub
- AMD Ryzen / Radeon GPU acceleration

Usage of AMD Products/Solutions

PersonaAI leverages AMD high-performance processors and GPUs to:

- Accelerate machine learning model training
- Enable real-time clustering inference
- Optimize parallel data processing
- Reduce latency in large-scale consumer systems

AMD hardware ensures scalable, energy-efficient, and high-speed AI execution suitable for real-time personalization engines.

Estimated implementation cost (optional)

Prototype Development:
- Development Tools – Open Source (Free)
- Cloud Hosting – ₹3,000 to ₹5,000
- Testing & Deployment – ₹2,000

Estimated Prototype Cost: ₹5,000 – ₹10,000

Enterprise deployment cost will vary based on scale.

Prototype Assets (Optional)

- ➔ GitHub Public Repository Link
- ➔ Demo Video Link (Max: 3 Minutes)

Add as per the requirements of the contest

- Focus Area: AI in Consumer Experiences
- Real-time adaptive personalization
- Scalable AI architecture
- Industry applicability
- AMD-accelerated computation
- Feasible prototype implementation

PersonaAI aims to redefine next-generation consumer personalization through intelligent behavioral analytics.

AMD Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by H2S HACK2SKILL

Thank you!